

DATA MINING LAB, 2026
Institute of Engineering, Thapathali Campus,
Tribhuvan University

Submit your jupyter notebook file for each labs work separately to the google drive link [DataMiningLab2083](#) . Create your own folder using your roll number (THA079BCT001 or THA079BEI010) and upload files inside your roll number folder. File name should be *lab1.ipynb*, *lab2.ipynb*, *lab3.ipynb*, *lab4.ipynb* and *lab5.ipynb*

Deadline: August 15, 2026 11:59 PM.

Lab 1: Python Basics

1. Write a program that takes three numbers and prints the largest one using if...elif...else.
2. Print the multiplication table of a given number using a for loop.
3. Write a program that keeps asking the user to enter a number until they enter -1, and then prints the sum of all entered numbers (excluding -1).
4. Given a list of tuples with student names and marks, write a program to print names of students who scored more than 80.
students = [("Amit", 85), ("Rita", 78), ("John", 90)]
5. Write a function is_prime(n) that returns True if a number is prime, else returns False.
6. Write a function is_palindrome(word) that returns True if the word is a palindrome (same forwards and backwards), and False otherwise.
7. Create a Python file named calculator.py that contains the following functions:
 - a. add(a, b)
 - b. subtract(a, b)
 - c. multiply(a, b)
 - d. divide(a, b)

Then, write another Python script that imports this module and calls each function with sample inputs.

Lab 2: Numpy & Matplotlib

1. Write a program to find the following between two vectors $a = [2,3,7,8,7]$ and $b = [4,7,8,2,1]$.
 - a. Euclidean distance
 - b. Manhattan distance
 - c. Cosine Similarity
2. Write a program to find a simple matching coefficient and Jaccard similarity between $[1,0,1,0,0,0]$ and $[1,0,1,1,0,1]$.
3. Write a program to perform feature scaling (min max, z score and decimal scaling) on $[35000, 42000, 39000, 45000]$
[1-3 using numpy only]
4. You are provided with the following arrays representing the monthly sales data (in USD) of two products — Laptops and Smartphones — sold by an electronics store over the first 6 months of the year:

`months = ['January', 'February', 'March', 'April', 'May', 'June']`

`laptop_sales = [12000, 15000, 17000, 13000, 16000, 18000]`

`smartphone_sales = [10000, 14000, 16000, 12000, 15500, 17500]`

- a. Plot the sales trends of both laptops and smartphones over the 6 months.
- b. Create a scatter plot comparing laptop sales vs smartphone sales.
- c. Draw a bar chart showing sales of laptops and smartphones for each month.
- d. Plot a pie chart showing the percentage contribution of each month to total laptop sales.
- e. Plot another pie chart for smartphone sales.

Lab 3: Pandas & Data Preprocessing

```
ecommerce_data = {  
    'OrderID': [101, 102, 103, 104, 105, 106, 107],  
    'CustomerName': ['Alice', 'Bob', 'Charlie', 'David', 'Eve', 'Frank', 'Grace'],  
    'Category': ['Electronics', 'Fashion', 'Electronics', 'Home', 'Fashion', 'Electronics',  
    'Home'],  
    'Amount': [200, 150, 300, 400, 250, 500, 350],  
    'Discount': [20, 10, 30, 50, 20, 60, 40],  
    'PurchaseDate': pd.to_datetime(['2023-11-20', '2023-11-21', '2023-11-22',  
    '2023-11-23', '2023-11-24', '2023-11-25', '2023-11-26']),  
    'City': ['New York', 'Los Angeles', 'New York', 'Chicago', 'Los Angeles', 'Chicago',  
    'New York']  
}
```

1. What is the shape of the dataset?
2. Are there any null values in the dataset?
3. Retrieve [Charlie Electronics 300] using loc and iloc
4. Fetch rows for orders placed by customers from New York.
5. What is the average order amount? What is the total discount provided?
6. Sort the orders by Amount in descending order.
7. Display all orders where the Category is "Electronics" and the Amount is greater than 250.

```
data = [  
    {"id": 1, "age": 25, "income": 50000, "city": "New York", "purchased": "yes"},  
    {"id": 2, "age": 30, "income": 60000, "city": "San Francisco", "purchased": "no"},  
    {"id": 3, "age": None, "income": 75000, "city": "New York", "purchased": None},  
    {"id": 4, "age": 22, "income": None, "city": "Los Angeles", "purchased": "yes"},  
    {"id": 5, "age": 35, "income": 80000, "city": None, "purchased": "no"},  
    {"id": 6, "age": 29, "income": 58000, "city": "San Francisco", "purchased": "yes"},  
    {"id": 7, "age": None, "income": None, "city": "New York", "purchased": "no"},  
    {"id": 8, "age": 40, "income": 90000, "city": "San Francisco", "purchased": "yes"},  
    {"id": 9, "age": 28, "income": 65000, "city": "Los Angeles", "purchased": None},  
    {"id": 10, "age": 33, "income": 72000, "city": "New York", "purchased": "no"},  
    {"id": 11, "age": None, "income": 50000, "city": "Los Angeles", "purchased": "yes"},  
    {"id": 12, "age": 26, "income": 61000, "city": None, "purchased": "yes"},  
]
```

```
{ "id": 13, "age": 45, "income": 95000, "city": "San Francisco", "purchased": None },  
{ "id": 14, "age": 50, "income": None, "city": "Los Angeles", "purchased": "no" },  
{ "id": 15, "age": 38, "income": 70000, "city": "New York", "purchased": "yes" },
```

]

1. Replace None in numerical fields (age, income) with mean/median.
2. Replace None in categorical fields (city, purchased) with mode or a placeholder like "unknown".
3. Normalize numerical fields (age, income) to a scale of 0-1 using min-max scaling or standardization (z-score).
4. Use one-hot encoding for the city.
5. Use label encoding (e.g., yes=1, no=0) for purchase.

Lab 4: KNN / Decision Trees

1. Create a confusion matrix (accuracy ,precision, recall, specificity,fpr, ,f1score) from scratch without using sk-learn. (use numpy)
2. Construct a Decision Tree and KNN Algorithm on the Haberman dataset and evaluate its performance using accuracy,precision, recall and f1 score.

Lab 5: Clustering

1. You are given the following 2D data points. Apply K-Means clustering and use the Elbow Method to determine the optimal number of clusters.
[1, 2], [2, 3], [3, 3], [10, 12], [11, 13], [12, 14],[25, 30], [26, 29], [27, 28]
2. Perform agglomerative hierarchical clustering on the following data points. Plot the dendrogram and identify the optimal number of clusters.
[1, 1], [2, 1], [1, 2], [8, 8], [9, 9], [8, 9], [4, 7], [5, 6]
3. Apply the DBSCAN algorithm to find core, border, and noise points from the following spatial dataset.
[1, 2], [1.1, 2.1], [0.9, 1.8], [1.2, 2.2], [8, 8], [8.1, 8.2], [8.2, 7.9], [5, 5]